福建理工大学
FUJIAN UNIVERSITY OF TECHNOLOGY

《计算机网络技术及应用》
实验五报告

申报学校：福建理工大学

学　　院：计算机科学与数学学院

课程类型：专业基础课

适用专业：网络空间安全 22

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实验五 动态路由 RIP

实验目的：

1) 阐释 VRP 的基本操作命令，并熟练配置

2) 解释 D-V 算法，阐释 RIP 工作原理

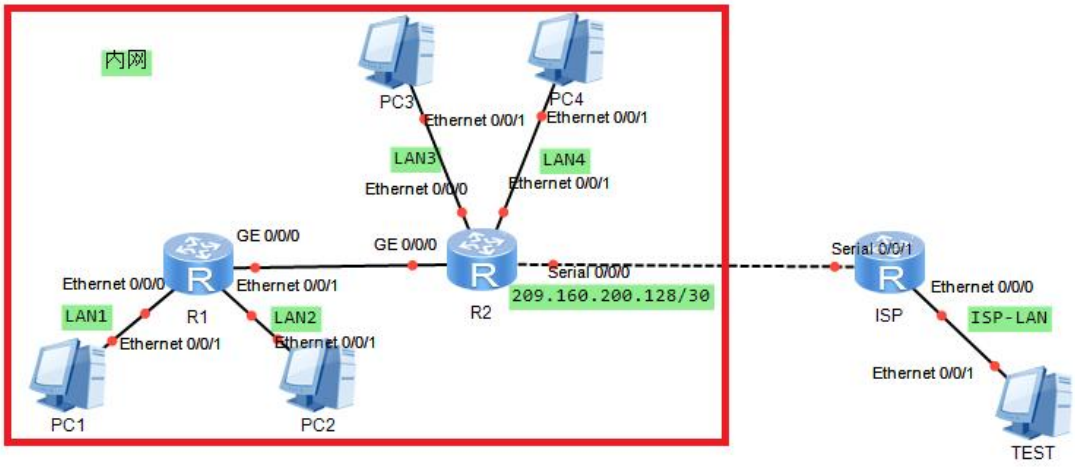
3) 解释反向发布路由的原因并熟练配置

4) 能解释 RIP1 和 RIP2 的区别，并分析如何通过实验验证

5) 分析内外网之间的通信

实验内容：

一、场景



场景说明：R1 和 R2 是内网路由器，通过 ISP 连接到 Internet，TEST 为因特网上的测试主机，该主机地址可以采用任意的外网地址,本实验忽略 NAT。

二、实验要求

1)使用 10.20.30.0/24 网络划分子网，LAN1 需要 30 个主机，LAN2 需要 12 个主机，LAN3 和 LAN4 各需要 50 个主机，为内网各个子网划分网络号，ISP-LAN 的地址可以自己定义任意外网地址。

2）每个子网的第 1 个有效地址分配给网关，最后一个有效地址分配给主机

3) 路由配置要求：在 R1,R2 上配置 rip 动态路由，R2 配置默认路由访问外网，R2 通过反向发布使得 R1 学习到默认路由，ISP 上配置总结路由。

4）连通要求：内网网所有的主机之间都可以访问

5）路由器改名要求：”学号-R1”，”学号-R2”，”学号-ISP”

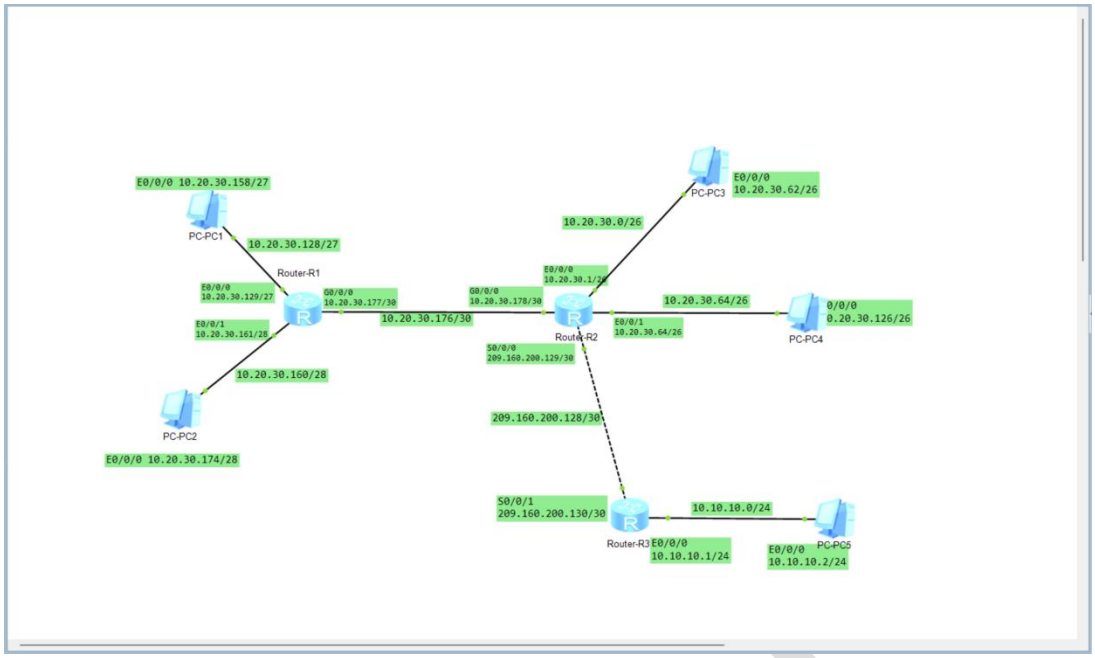
说明：如果是在实验四直接修改，请将所有和静态路由相关的路由先去掉。

三、实验步骤

1、子网划分，填写地址表

3-1 IP 地址表				
设备	**接口**	**IP 地址**	**子网掩码**	**默认网关**
R1	E0/0/0	10.20.30.129	255.255.255.224	无
	E0/0/1	10.20.30.161	255.255.255.240	无
	G0/0/0	10.20.30.177	255.255.255.252	无
R2	E0/0/0	10.20.30.1	255.255.255.192	无
	E0/0/1	10.20.30.65	255.255.255.192	无
	G0/0/0	10.20.30.178	255.255.255.252	无
	S0/0/0	209.160.200.129	255.255.255.252	无
ISP	E0/0/0	10.10.10.1	255.255.255.0	无
	S0/0/1	209.160.200.130	255.255.255.252	无
PC1	网卡	10.20.30.158	255.255.255.224	10.20.30.129
PC2	网卡	10.20.30.174	255.255.255.240	10.20.30.161
PC3	网卡	10.20.30.62	255.255.255.192	10.20.30.1
PC4	网卡	10.20.30.126	255.255.255.192	10.20.30.65
TEST	网卡	10.10.10.2	255.255.255.0	10.10.10.1

2、配置每个路由器和主机的 IP 地址**拓扑：**



R1：

```
<Huawei>sy
[Huawei]sysname 3220812128-R1
[3220812128-R1]int g0/0/0
[3220812128-R1-GigabitEthernet0/0/0]ip add 10.20.30.177 30
[3220812128-R1-GigabitEthernet0/0/0]int e0/0/0
[3220812128-R1-Ethernet0/0/0]ip add 10.20.30.129 27
[3220812128-R1-Ethernet0/0/0]int e0/0/1
[3220812128-R1-Ethernet0/0/1]ip add 10.20.30.161 28
[3220812128-R1-Ethernet0/0/1]q
[3220812128-R1]ip route-static 0.0.0.0 0 10.20.30.178
```

R2:

```
<Huawei>sy
[Huawei]sysname 3220812128-R2
[3220812128-R2]int g0/0/0
[3220812128-R2-GigabitEthernet0/0/0]ip add 10.20.30.178 30
[3220812128-R2-GigabitEthernet0/0/0]int e0/0/0
[3220812128-R2-Ethernet0/0/0]ip add 10.20.30.1 26
[3220812128-R2-Ethernet0/0/0]int e0/0/1
[3220812128-R2-Ethernet0/0/1]ip add 10.20.30.65 26
[3220812128-R2-Ethernet0/0/1]int s0/0/0
[3220812128-R2-Serial0/0/0]ip add 209.160.200.129 30
[3220812128-R2-Serial0/0/0]q
[3220812128-R2]ip route-static 0.0.0.0 0 209.160.200.130
[3220812128-R2]ip route-static 10.20.30.128 27 10.20.30.177
[3220812128-R2]ip route-static 10.20.30.160 28 10.20.30.177
```

ISP:

```
<Huawei>sy
[Huawei]sysname 3220812128-ISP
[3220812128-ISP]int e0/0/0
[3220812128-ISP-Ethernet0/0/0]ip add 10.10.10.1 24
[3220812128-ISP-Ethernet0/0/0]int s0/0/1
[3220812128-ISP-Serial0/0/1]ip add 209.160.200.130 30
[3220812128-ISP-Serial0/0/1]q
[3220812128-ISP]ip route-static 10.20.30.0 24 209.160.200.129
```

3、检查地址配置

R1：

```
[3220812128-R1]dis ip int brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 4
The number of interface that is DOWN in Physical is 7
The number of interface that is UP in Protocol is 4
The number of interface that is DOWN in Protocol is 7

Interface                         IP Address/Mask      Physical   Protocol
Ethernet0/0/0                     10.20.30.129/27      up         up
Ethernet0/0/1                     10.20.30.161/28      up         up
GigabitEthernet0/0/0              10.20.30.177/30      up         up
GigabitEthernet0/0/1              unassigned           down       down
GigabitEthernet0/0/2              unassigned           down       down
GigabitEthernet0/0/3              unassigned           down       down
NULL0                             unassigned           up         up(s)
Serial0/0/0                       unassigned           down       down
Serial0/0/1                       unassigned           down       down
Serial0/0/2                       unassigned           down       down
Serial0/0/3                       unassigned           down       down
```

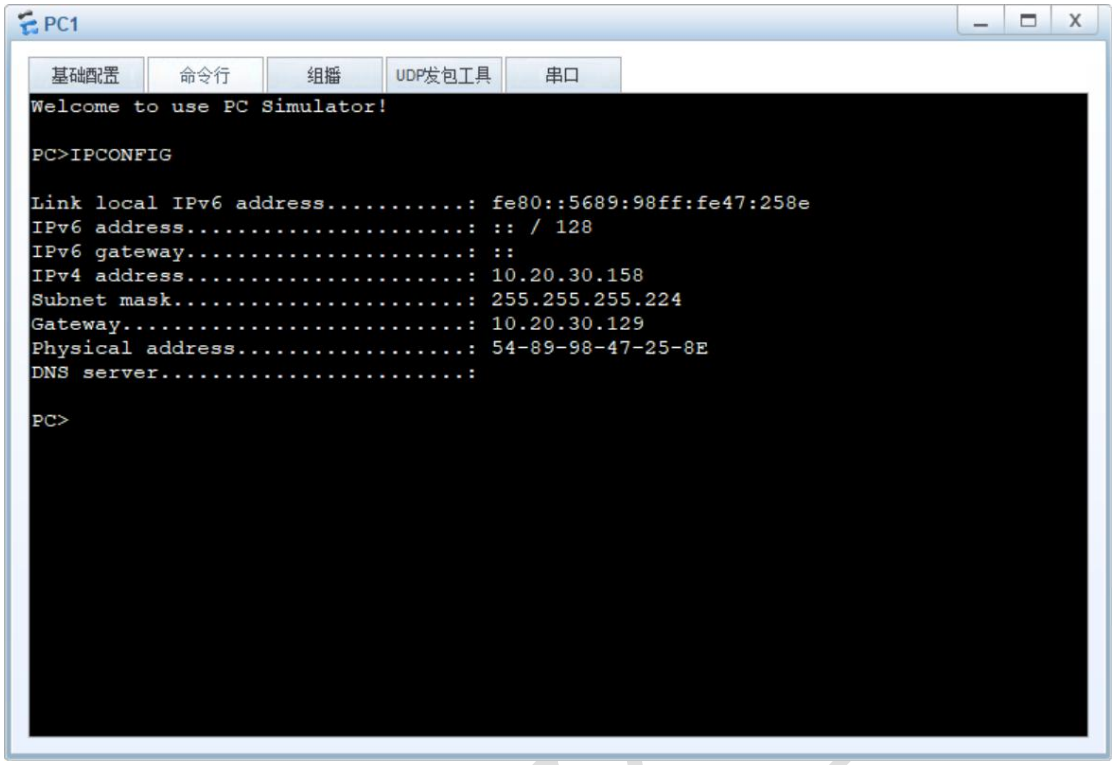
R2:

```
[3220812128-R2]dis ip int b
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 5
The number of interface that is DOWN in Physical is 6
The number of interface that is UP in Protocol is 5
The number of interface that is DOWN in Protocol is 6

Interface                         IP Address/Mask      Physical   Protocol
Ethernet0/0/0                     10.20.30.1/26        up         up
Ethernet0/0/1                     10.20.30.65/26       up         up
GigabitEthernet0/0/0              10.20.30.178/30      up         up
GigabitEthernet0/0/1              unassigned           down       down
GigabitEthernet0/0/2              unassigned           down       down
GigabitEthernet0/0/3              unassigned           down       down
NULL0                             unassigned           up         up(s)
Serial0/0/0                       209.160.200.129/30   up         up
Serial0/0/1                       unassigned           down       down
Serial0/0/2                       unassigned           down       down
Serial0/0/3                       unassigned           down       down
```

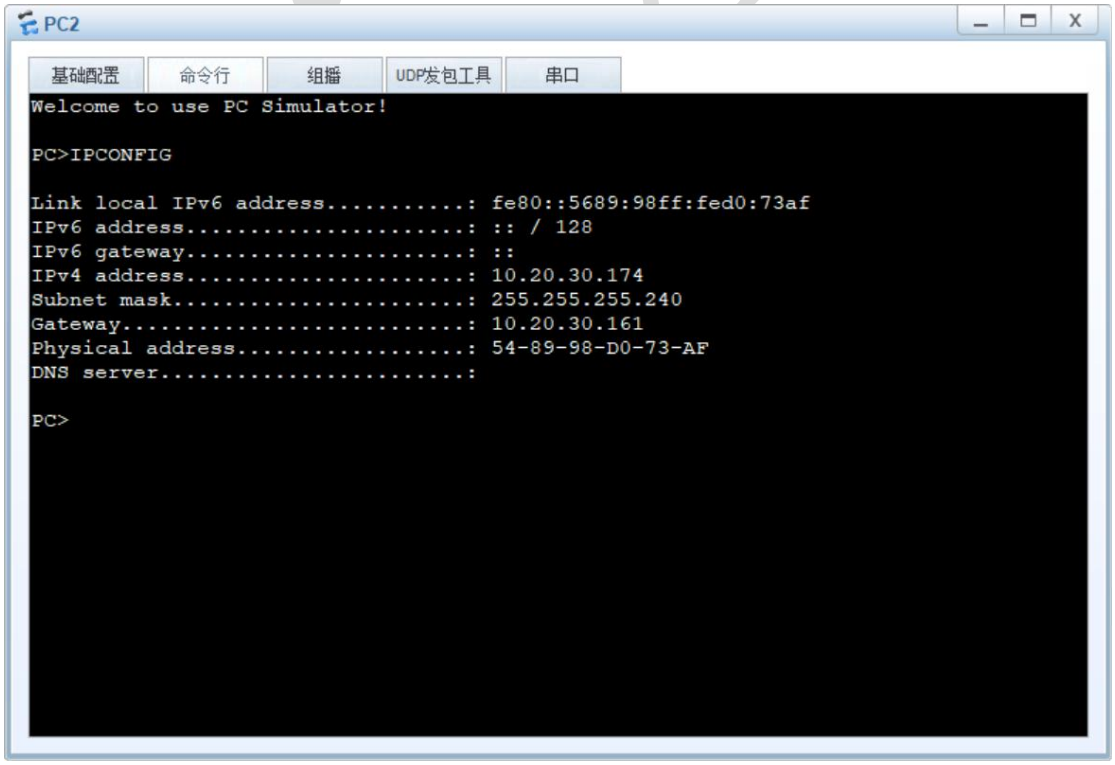
ISP:

```
[3220812128-ISP]dis ip int b
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 3
The number of interface that is DOWN in Physical is 8
The number of interface that is UP in Protocol is 3
The number of interface that is DOWN in Protocol is 8

Interface                         IP Address/Mask      Physical   Protocol
Ethernet0/0/0                     10.10.10.1/24        up         up
Ethernet0/0/1                     unassigned           down       down
GigabitEthernet0/0/0              unassigned           down       down
GigabitEthernet0/0/1              unassigned           down       down
GigabitEthernet0/0/2              unassigned           down       down
GigabitEthernet0/0/3              unassigned           down       down
NULL0                             unassigned           up         up(s)
Serial0/0/0                       unassigned           down       down
Serial0/0/1                       209.160.200.130/30   up         up
Serial0/0/2                       unassigned           down       down
Serial0/0/3                       unassigned           down       down
```

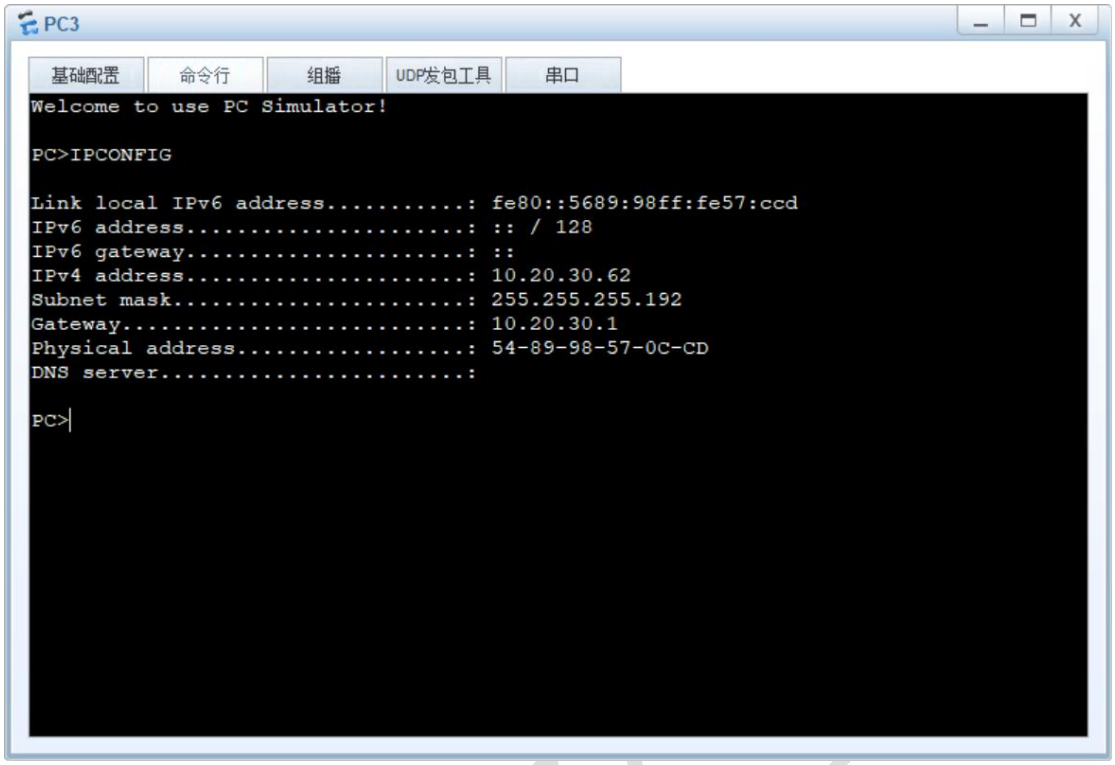
PC1:


PC1
基础配置
命令行
组播
UDP发包工具
串口
Welcome to use PC Simulator!
PC>IPCONFIG
Link local IPv6 address.....: fe80::5689:98ff:fe47:258e
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.20.30.158
Subnet mask.....: 255.255.255.224
Gateway.....: 10.20.30.129
Physical address.....: 54-89-98-47-25-8E
DNS server.....:
PC>

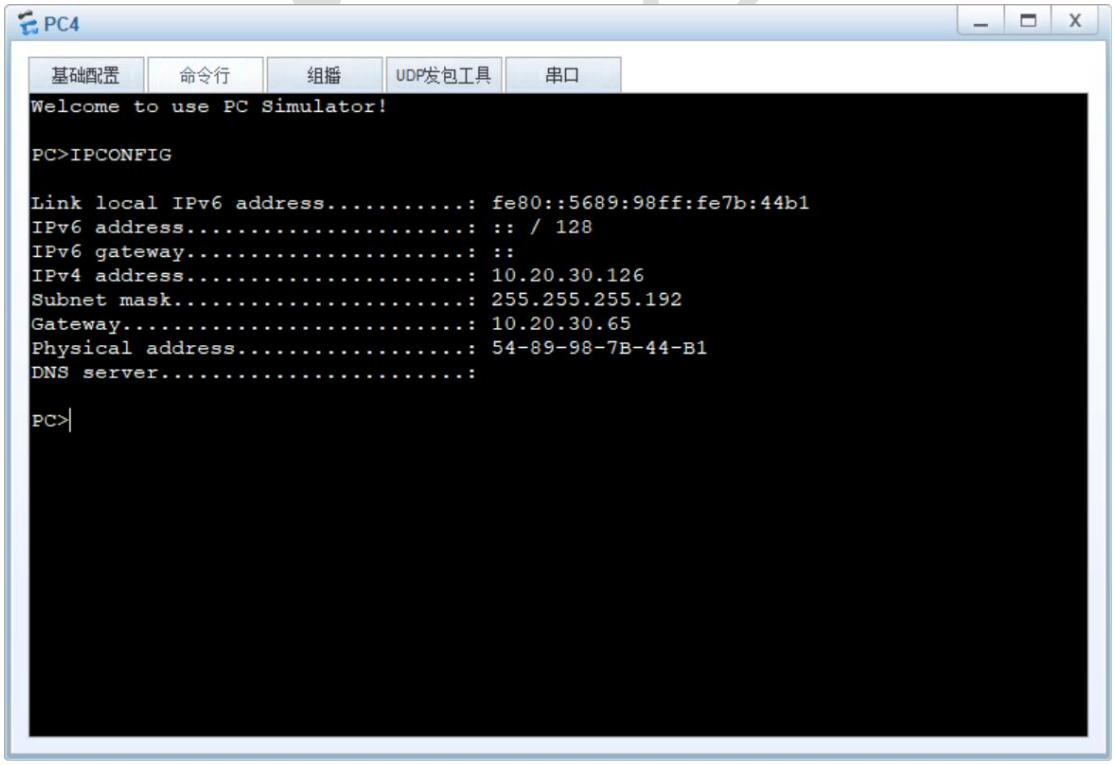
PC2:


PC2
基础配置
命令行
组播
UDP发包工具
串口
Welcome to use PC Simulator!
PC>IPCONFIG
Link local IPv6 address.....: fe80::5689:98ff:fed0:73af
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.20.30.174
Subnet mask.....: 255.255.255.240
Gateway.....: 10.20.30.161
Physical address.....: 54-89-98-D0-73-AF
DNS server.....:
PC>

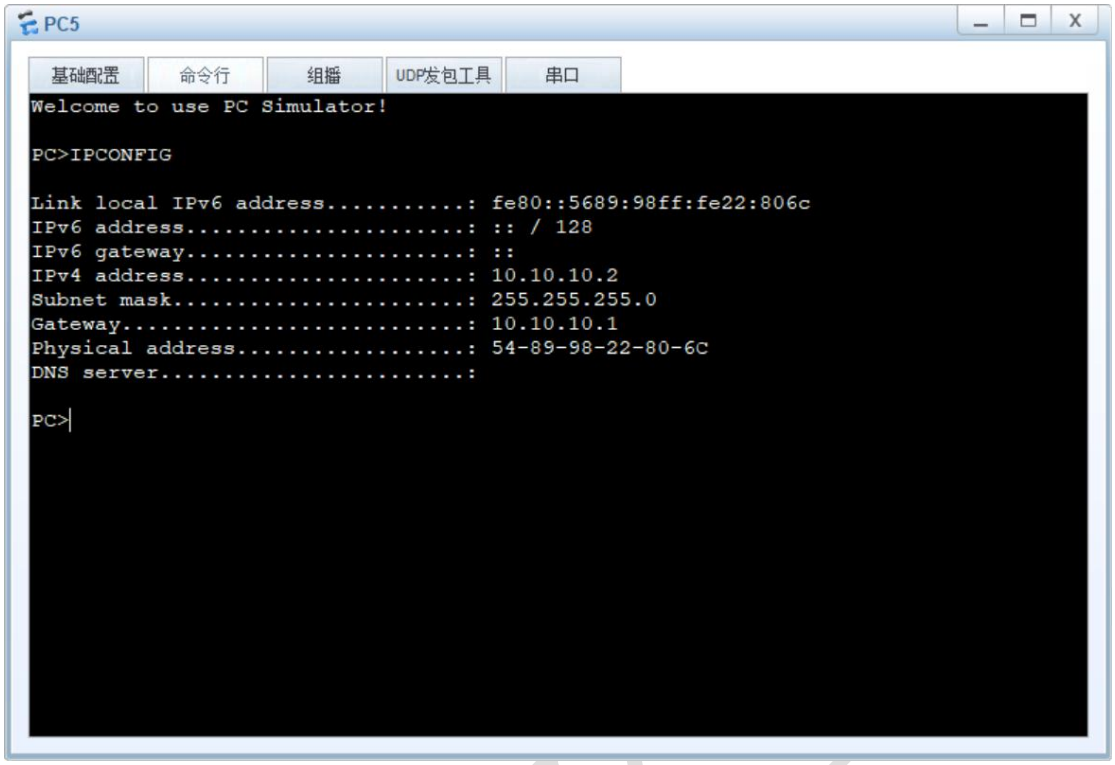
PC3:


PC3
基础配置
命令行
组播
UDP发包工具
串口
Welcome to use PC Simulator!
PC>IPCONFIG
Link local IPv6 address.....: fe80::5689:98ff:fe57:ccd
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.20.30.62
Subnet mask.....: 255.255.255.192
Gateway.....: 10.20.30.1
Physical address.....: 54-89-98-57-0C-CD
DNS server.....:
PC>

PC4:

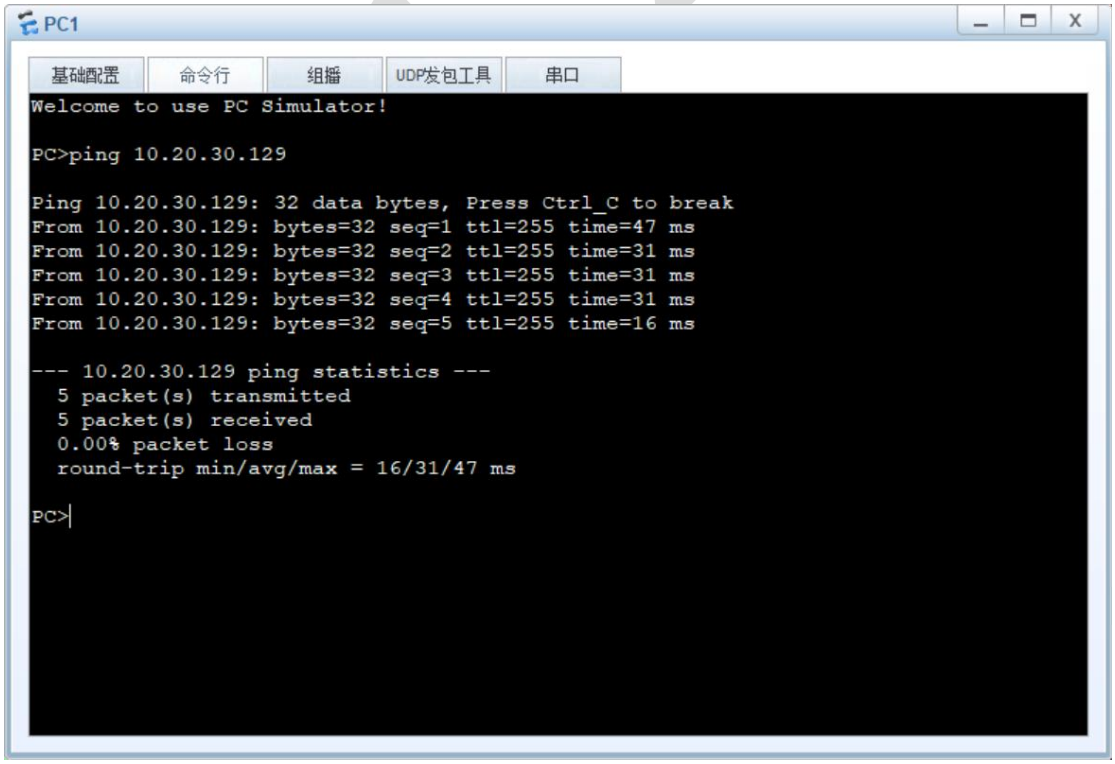

PC4
基础配置
命令行
组播
UDP发包工具
串口
Welcome to use PC Simulator!
PC>IPCONFIG
Link local IPv6 address.....: fe80::5689:98ff:fe7b:44b1
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.20.30.126
Subnet mask.....: 255.255.255.192
Gateway.....: 10.20.30.65
Physical address.....: 54-89-98-7B-44-B1
DNS server.....:
PC>

PC5:

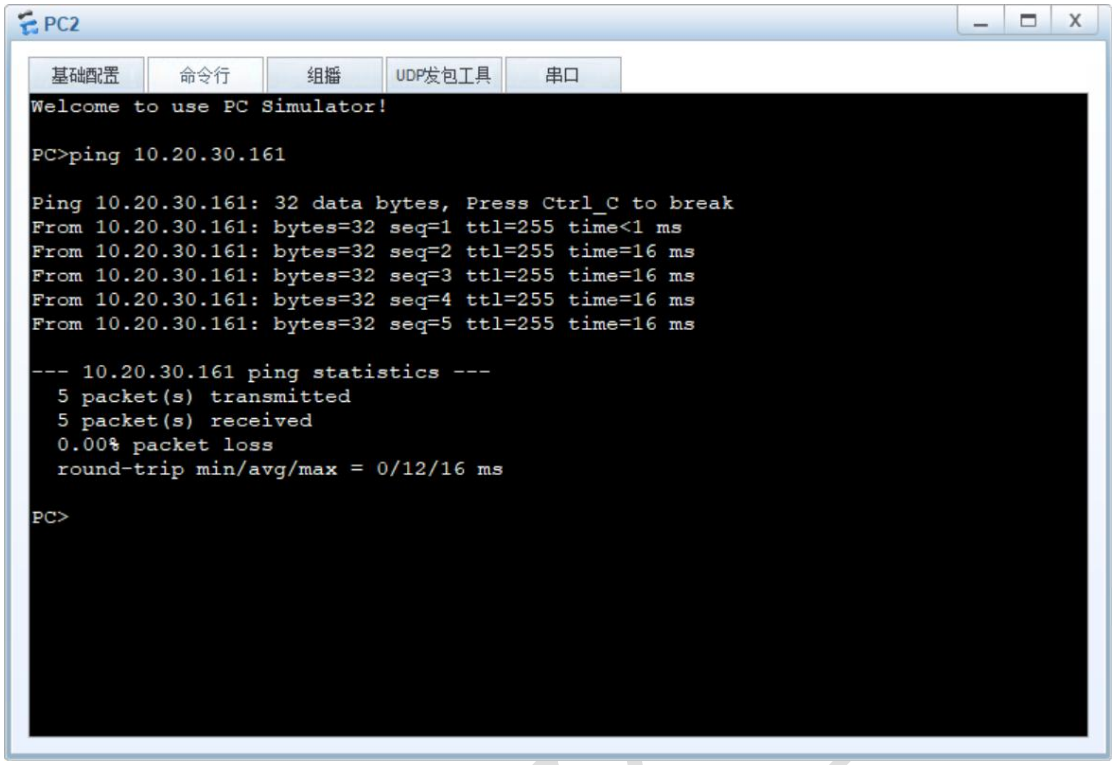


4、测试主机与网关，路由器之间的连通性，如果有问题请先排错

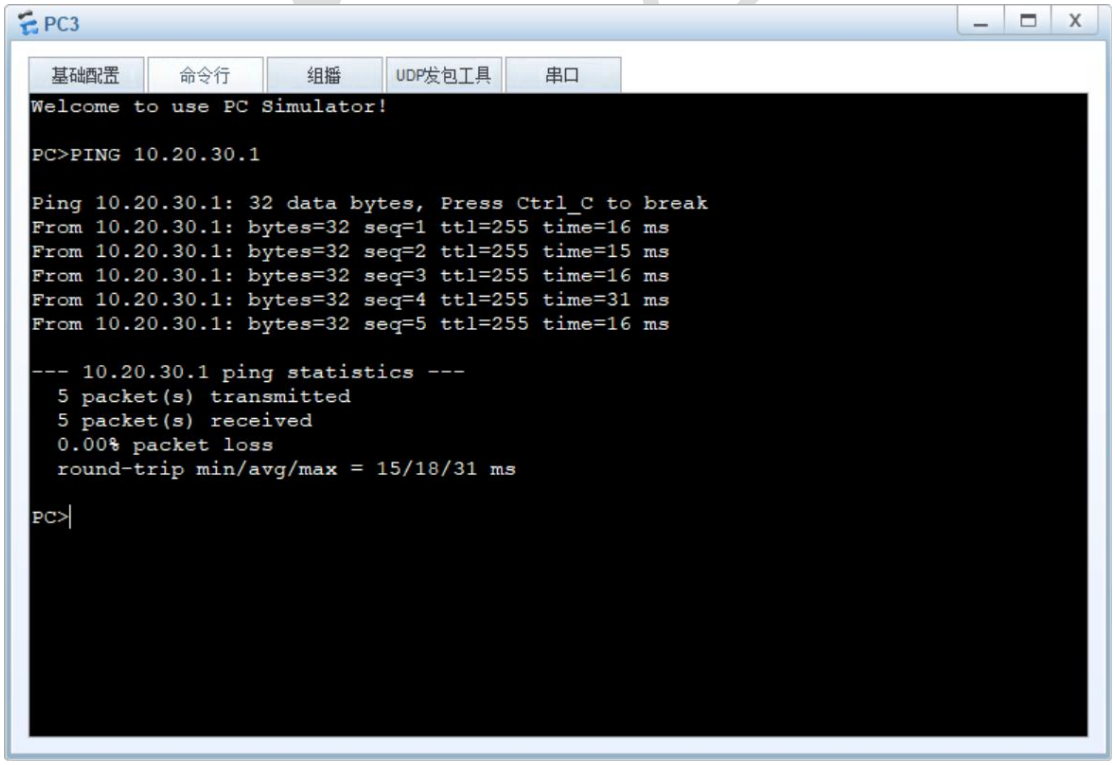
PC1 ping 网关：



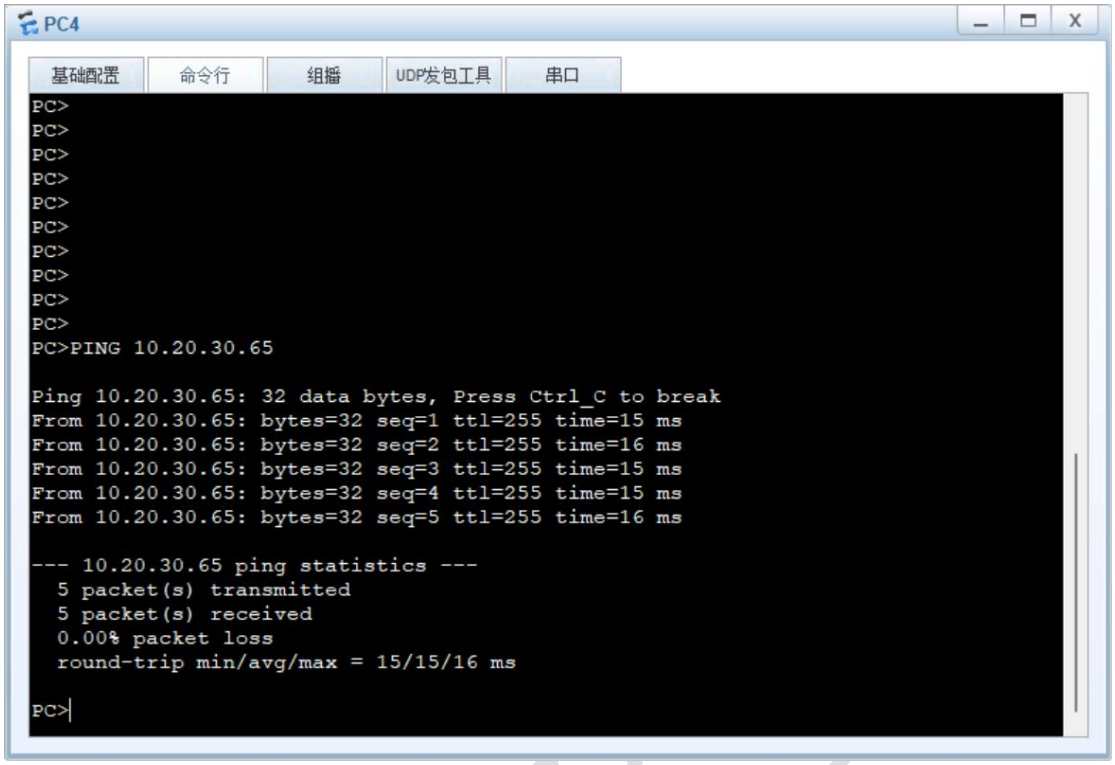
PC2 ping 网关：



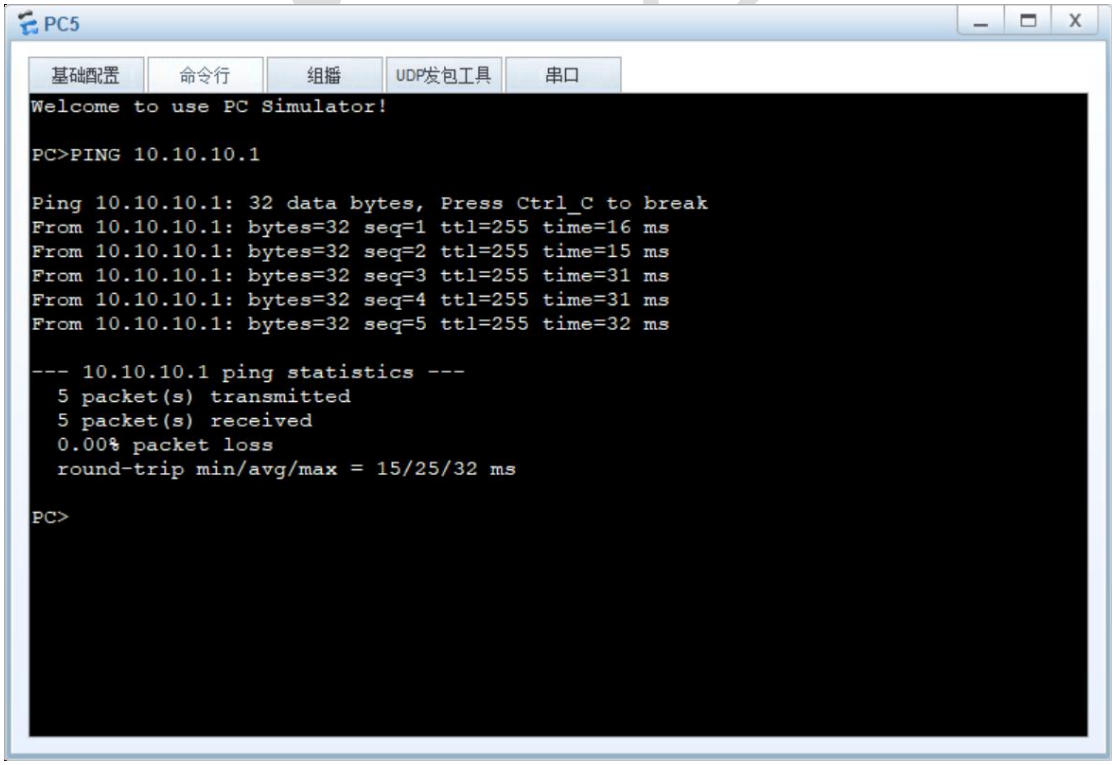
PC3 ping 网关:



PC4 ping 网关:



TEST ping 网关:



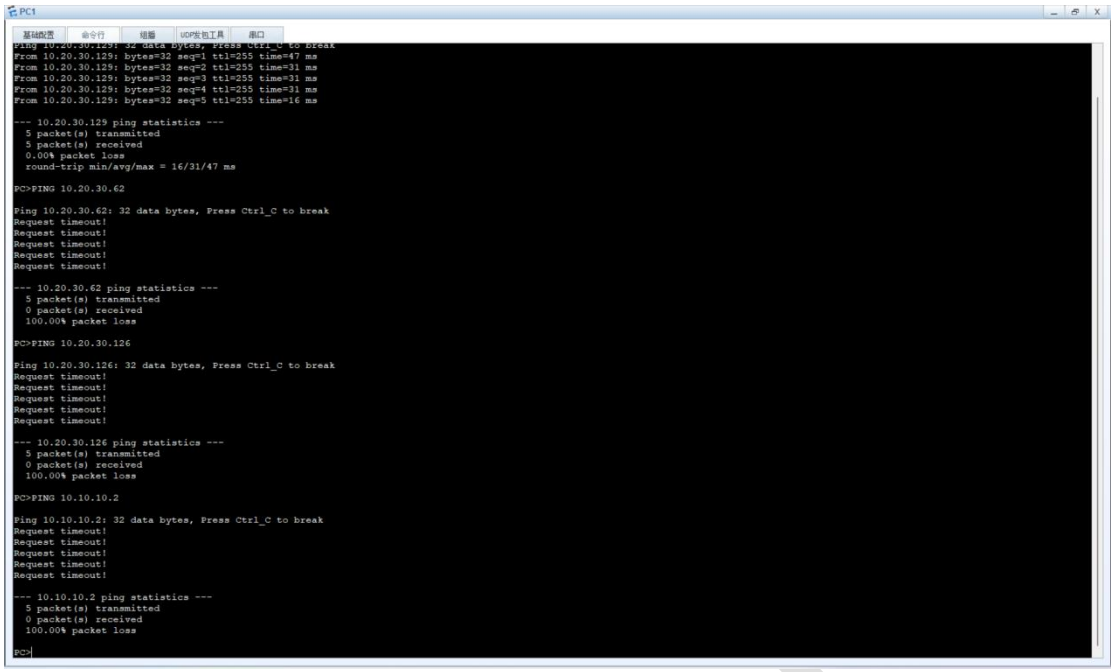
R1 PING R2:

```
R1
[3220812128-R1]undo ip rd-filter
[3220812128-R1]undo ip route-static all
Warning: This operation may lead to the deletion of all the public IPv4 static
outes and their configurations. Continue? [Y/N]:y
[3220812128-R1]
May 15 2024 09:45:35-08:00 3220812128-R1 %%01RM/4/IPV4_DEFT_RT_CHG(l)[0]:IPV4 d
fault Route is changed. (ChangeType=Delete,InstanceId=0,Protocol=Static,ExitIf=
igabitEthernet0/0/0,Nexthop=10.20.30.178,Neighbour=0.0.0.0,Preference=60,Label=
ULL,Metric=0)
<3220812128-R1>PING 10.20.30.178
  PING 10.20.30.178: 56  data bytes, press CTRL_C to break
    Reply from 10.20.30.178: bytes=56 Sequence=1 ttl=255 time=90 ms
    Reply from 10.20.30.178: bytes=56 Sequence=2 ttl=255 time=30 ms
    Reply from 10.20.30.178: bytes=56 Sequence=3 ttl=255 time=20 ms
    Reply from 10.20.30.178: bytes=56 Sequence=4 ttl=255 time=40 ms
    Reply from 10.20.30.178: bytes=56 Sequence=5 ttl=255 time=30 ms

  --- 10.20.30.178 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/42/90 ms

<3220812128-R1>
```

R2 PING ISP:

```
R2
The device is running!

<3220812128-R2>sy
Enter system view, return user view with Ctrl+Z.
[3220812128-R2]undo ip route-static all -y
                                        ^
Error:Too many parameters found at '^' position.
[3220812128-R2]undo ip route-static all
Warning: This operation may lead to the deletion of all the public IPv4 static r
outes and their configurations. Continue? [Y/N]:y
[3220812128-R2]
<3220812128-R2>PING 209.160.200.130
  PING 209.160.200.130: 56  data bytes, press CTRL_C to break
    Reply from 209.160.200.130: bytes=56 Sequence=1 ttl=255 time=30 ms
    Reply from 209.160.200.130: bytes=56 Sequence=2 ttl=255 time=40 ms
    Reply from 209.160.200.130: bytes=56 Sequence=3 ttl=255 time=30 ms
    Reply from 209.160.200.130: bytes=56 Sequence=4 ttl=255 time=20 ms
    Reply from 209.160.200.130: bytes=56 Sequence=5 ttl=255 time=10 ms

  --- 209.160.200.130 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 10/26/40 ms

<3220812128-R2>
```

5、测试 PC1 与任何 PC3\PC4\TEST 任何一台主机之间的连通性



无法连通·

6、R1 和 R2 路由器配置 RIPv1

R1：

```
<3220812128-R1>sy
[3220812128-R1]rip 1
[3220812128-R1-rip-1]version 1
[3220812128-R1-rip-1]network 10.0.0.0
[3220812128-R1-rip-1]
```

R2:

```
<3220812128-R2>sy
[3220812128-R2]rip 1
[3220812128-R2-rip-1]version 1
[3220812128-R2-rip-1]network 10.0.0.0
[3220812128-R2-rip-1]
```

7、查看两个路由器配置后的结果，并分析

R1：

```
[3220812128-R1]dis ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 9        Routes : 9

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

       10.0.0.0/8   RIP     100  1           D   10.20.30.178    GigabitEthernet
0/0/0
   10.20.30.128/27  Direct  0    0           D   10.20.30.129    Ethernet0/0/0
   10.20.30.129/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/0
   10.20.30.160/28  Direct  0    0           D   10.20.30.161    Ethernet0/0/1
   10.20.30.161/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/1
   10.20.30.176/30  Direct  0    0           D   10.20.30.177    GigabitEthernet
0/0/0
   10.20.30.177/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[3220812128-R1]
```

R2:

```
[3220812128-R2]dis ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 13       Routes : 13

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

       10.0.0.0/8   RIP     100  1           D   10.20.30.177    GigabitEthernet
0/0/0
     10.20.30.0/26  Direct  0    0           D   10.20.30.1      Ethernet0/0/0
     10.20.30.1/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/0
    10.20.30.64/26  Direct  0    0           D   10.20.30.65     Ethernet0/0/1
    10.20.30.65/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/1
   10.20.30.176/30  Direct  0    0           D   10.20.30.178    GigabitEthernet
0/0/0
   10.20.30.178/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
209.160.200.128/30  Direct  0    0           D   209.160.200.129 Serial0/0/0
209.160.200.129/32  Direct  0    0           D   127.0.0.1       Serial0/0/0
209.160.200.130/32  Direct  0    0           D   209.160.200.130 Serial0/0/0

[3220812128-R2]q
```

R1 和 R2 的路由表中都多出了一条 10.0.0.0/8 rip 的路由条目。

8、R1 和 R2 路由器配置 RIPv2

R1:

```
<3220812128-R1>sy
[3220812128-R1]RIP 1
[3220812128-R1-rip-1]version 2
[3220812128-R1-rip-1]network 10.0.0.0
```

R2:

```
<3220812128-R2>sy
[3220812128-R2]rip 1
[3220812128-R2-rip-1]version 2
[3220812128-R2-rip-1]version 2
[3220812128-R2-rip-1]network 10.0.0.0
[3220812128-R2-rip-1]q
```

9、查看两个路由器配置后的结果，对比步骤 7 的路由表分析

R1:

```
[3220812128-R1]dis ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 10       Routes : 10

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

     10.20.30.0/26  RIP     100  1           D   10.20.30.178    GigabitEthernet
0/0/0
    10.20.30.64/26  RIP     100  1           D   10.20.30.178    GigabitEthernet
0/0/0
   10.20.30.128/27  Direct  0    0           D   10.20.30.129    Ethernet0/0/0
   10.20.30.129/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/0
   10.20.30.160/28  Direct  0    0           D   10.20.30.161    Ethernet0/0/1
   10.20.30.161/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/1
   10.20.30.176/30  Direct  0    0           D   10.20.30.177    GigabitEthernet
0/0/0
   10.20.30.177/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[3220812128-R1]
```

R2:

```
[3220812128-R2]dis ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 14       Routes : 14

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

     10.20.30.0/26  Direct  0    0           D   10.20.30.1      Ethernet0/0/0
     10.20.30.1/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/0
    10.20.30.64/26  Direct  0    0           D   10.20.30.65     Ethernet0/0/1
    10.20.30.65/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/1
   10.20.30.128/27  RIP     100  1           D   10.20.30.177    GigabitEthernet
0/0/0
   10.20.30.160/28  RIP     100  1           D   10.20.30.177    GigabitEthernet
0/0/0
   10.20.30.176/30  Direct  0    0           D   10.20.30.178    GigabitEthernet
0/0/0
   10.20.30.178/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
209.160.200.128/30  Direct  0    0           D   209.160.200.129 Serial0/0/0
209.160.200.129/32  Direct  0    0           D   127.0.0.1       Serial0/0/0
209.160.200.130/32  Direct  0    0           D   209.160.200.130 Serial0/0/0

[3220812128-R2]
```

对比 ripv1 去获得的路由表来说，ripv2 所获得的路由表更细致，ripv1 只能得到 10.0.0.0/8 的网段，而 ripv2 可以精确到每个子网的网段例如 10.20.30.128 就是 LAN1 的网段。

10、R2 上配置默认路由并反向发布

```
[3220812128-R2]ip route-static 0.0.0.0 0 209.160.200.130
[3220812128-R2]rip 1
[3220812128-R2-rip-1]default-route originate
[3220812128-R2-rip-1]
```

11、查看 R1 上是否学习到默认路由，如果没有，请排错

```
R1
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 11       Routes : 11

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

        0.0.0.0/0   RIP     100  1           D   10.20.30.178    GigabitEtherne
0/0/0
     10.20.30.0/26  RIP     100  1           D   10.20.30.178    GigabitEtherne
0/0/0
    10.20.30.64/26  RIP     100  1           D   10.20.30.178    GigabitEtherne
0/0/0
   10.20.30.128/27  Direct  0    0           D   10.20.30.129    Ethernet0/0/0
   10.20.30.129/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/0
   10.20.30.160/28  Direct  0    0           D   10.20.30.161    Ethernet0/0/1
   10.20.30.161/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/1
   10.20.30.176/30  Direct  0    0           D   10.20.30.177    GigabitEtherne
0/0/0
   10.20.30.177/32  Direct  0    0           D   127.0.0.1       GigabitEtherne
0/0/0
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0

[3220812128-R1]
```

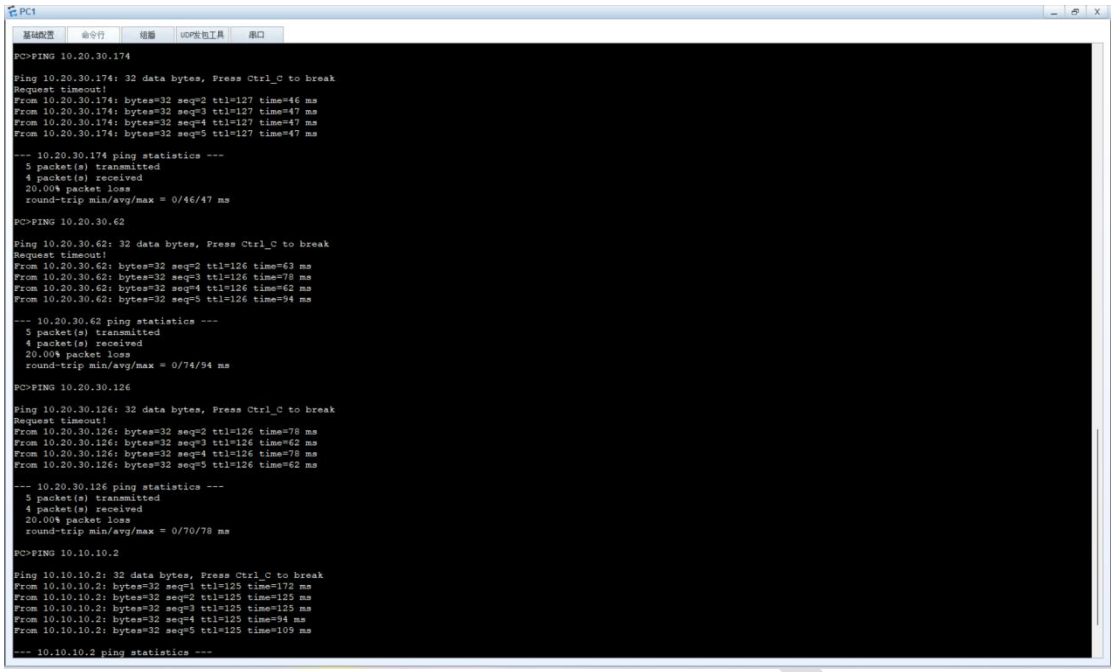
成功学习到了默认路由

12、ISP 上配置总结路由，并查看路由表

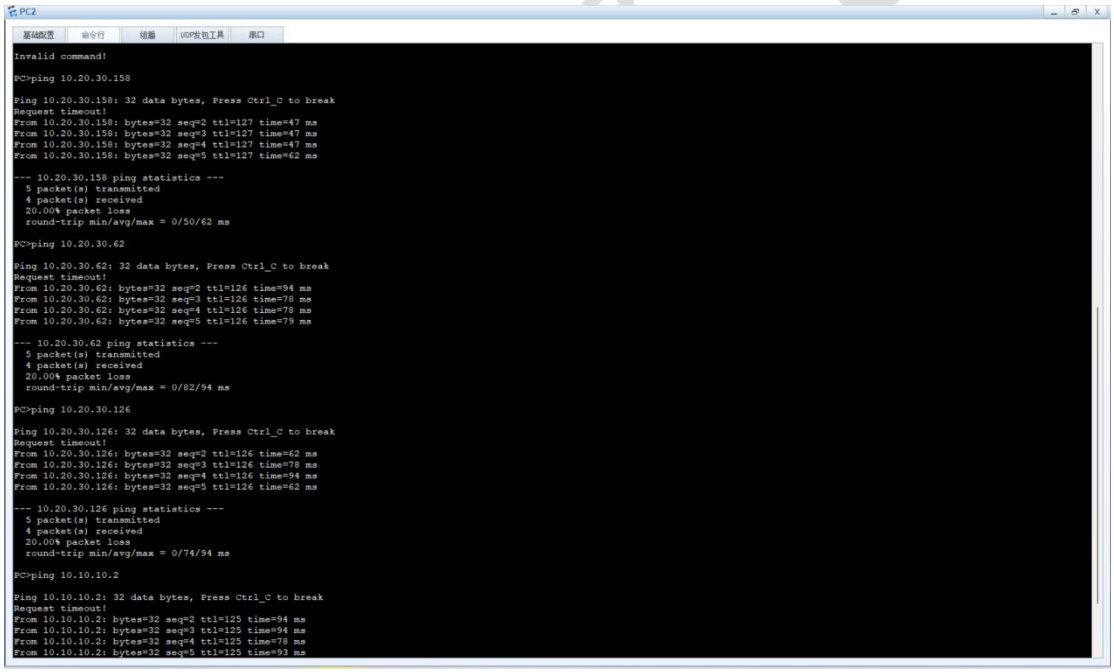
```
<3220812128-ISP>sy
[3220812128-ISP]ip route-static 10.20.30.0 24 209.160.200.129
[3220812128-ISP]dis ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 8        Routes : 8

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

     10.10.10.0/24  Direct  0    0           D   10.10.10.1      Ethernet0/0/0
     10.10.10.1/32  Direct  0    0           D   127.0.0.1       Ethernet0/0/0
     10.20.30.0/24  Static  60   0          RD   209.160.200.129 Serial0/0/1
      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
209.160.200.128/30  Direct  0    0           D   209.160.200.130 Serial0/0/1
209.160.200.129/32  Direct  0    0           D   209.160.200.129 Serial0/0/1
209.160.200.130/32  Direct  0    0           D   127.0.0.1       Serial0/0/1

[3220812128-ISP]
```

13、测试所有主机的连通性，有问题，请排错

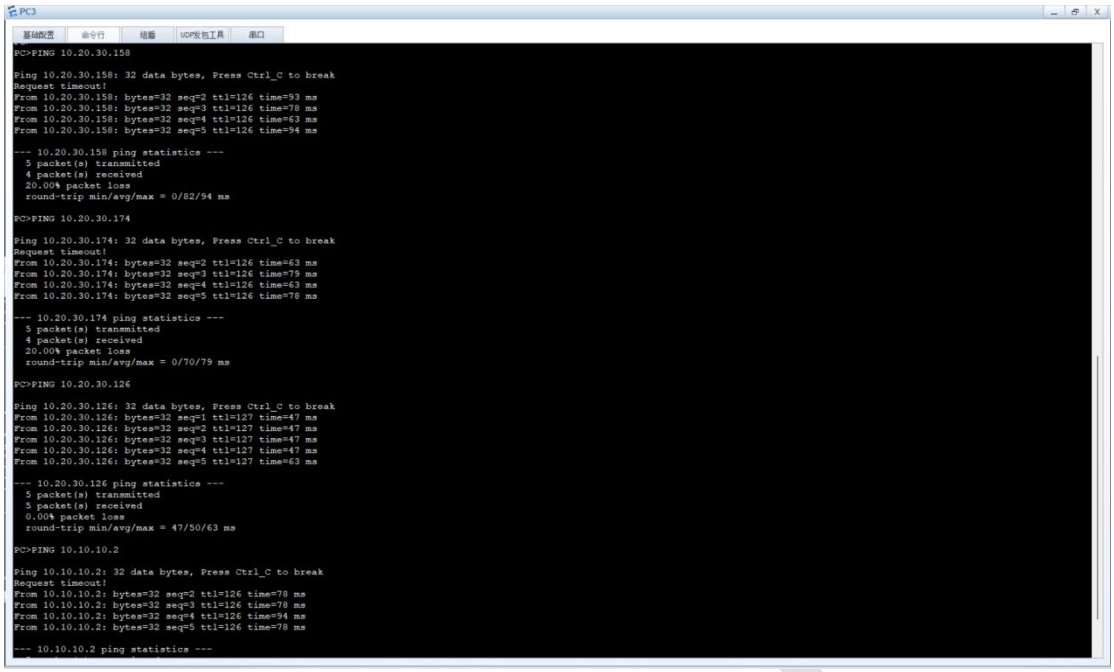
PC1 分别 ping PC2、PC3、PC4、TEST:



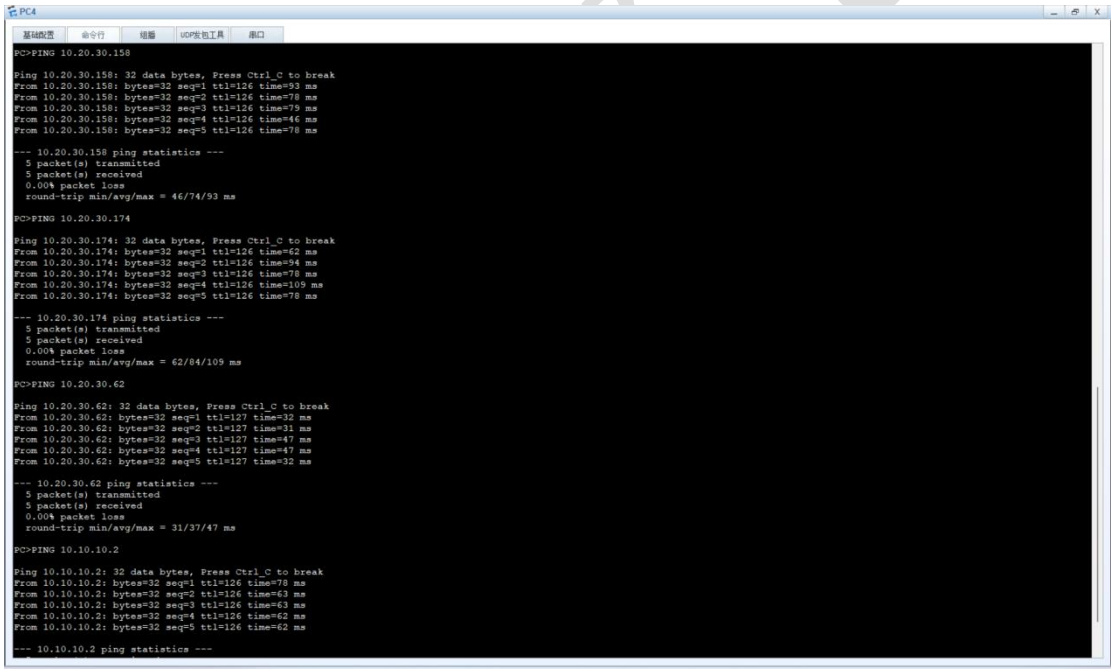
PC2 分别 ping PC1、PC3、PC4、TEST:



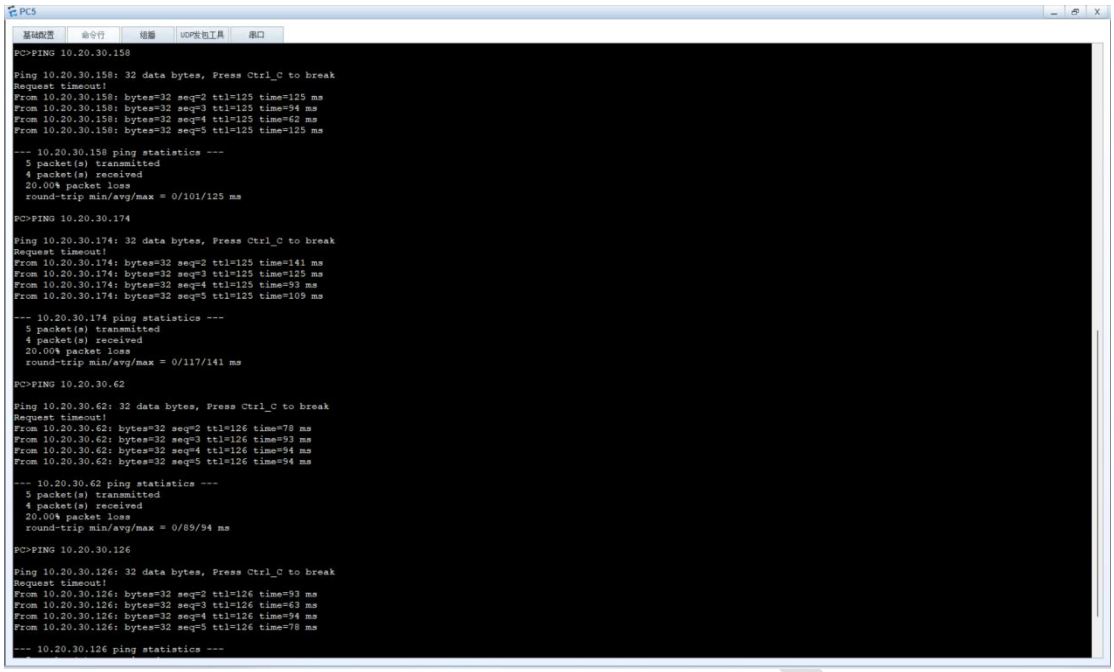
PC3 分别 ping PC1、PC2、PC4、TEST:



PC4 分别 ping PC1、PC2、PC3、TEST:



TEST 分别 ping PC1、PC2、PC3、PC4:



14、利用 debug 命令总结 RIPV1 和 RIPV2 的异同点

RIPV1:

```
May 15 2024 13:46:45.790.1-08:00 3220812128-R1 RIP/7/DBG: 6: 12176: RIP 1: Sendi
ng v1 response on GigabitEthernet0/0/0 from 10.20.30.177 with 3 RTEs
May 15 2024 13:46:45.790.2-08:00 3220812128-R1 RIP/7/DBG: 6: 12227: RIP 1: Sendi
ng response on interface GigabitEthernet0/0/0 from 10.20.30.177 to 255.255.255.2
55
May 15 2024 13:46:45.790.3-08:00 3220812128-R1 RIP/7/DBG: 6: 12247: Packet: Vers
ion 1, Cmd response, Length 64
May 15 2024 13:46:45.790.4-08:00 3220812128-R1 RIP/7/DBG: 6: 12296: Dest 0.0.0.0
, Cost 1
May 15 2024 13:46:45.790.5-08:00 3220812128-R1 RIP/7/DBG: 6: 12296: Dest 10.20.3
0.176, Cost 1
May 15 2024 13:46:45.790.6-08:00 3220812128-R1 RIP/7/DBG: 6: 12296: Dest 10.0.0.
0, Cost 1
May 15 2024 13:46:40.770.1-08:00 3220812128-R1 RIP/7/DBG: 6: 12185: RIP 1: Recei
ving v2 response on GigabitEthernet0/0/0 from 10.20.30.178 with 2 RTEs
May 15 2024 13:46:40.770.2-08:00 3220812128-R1 RIP/7/DBG: 6: 12236: RIP 1: Recei
ve response from 10.20.30.178 on GigabitEthernet0/0/0
May 15 2024 13:46:40.770.3-08:00 3220812128-R1 RIP/7/DBG: 6: 12247: Packet: Vers
ion 2, Cmd response, Length 44
May 15 2024 13:46:40.770.4-08:00 3220812128-R1 RIP/7/DBG: 6: 12315: Dest 10.20.3
0.0/26, Nexthop 0.0.0.0, Cost 1, Tag 0
May 15 2024 13:46:40.770.5-08:00 3220812128-R1 RIP/7/DBG: 6: 12315: Dest 10.20.3
0.64/26, Nexthop 0.0.0.0, Cost 1, Tag 0
May 15 2024 13:46:40.770.6-08:00 3220812128-R1 RIP/7/DBG: 6: 2500: RIP 1: Ignori
ng packet. This version is not configured.
May 15 2024 13:46:44.590.1-08:00 3220812128-R1 RIP/7/DBG: 25: 3741: RIP 1: Perio
dic timer expired for interface GigabitEthernet0/0/0
```

RIPV2:

相同点:

都是 RIP 协议，在大体上具有相同的性质，包括都是距离矢量算法，更新周期为 30 秒。

不同点:

RIPV1 不支持 VLSM，所以在内网中不是很精确。而 RIPV2 能够支持 VLSM，能够更准确的区分不同的内网子网。

在 RIPV1 中，它是用广播地址去发送

```
May 15 2024 13:46:45.790.1-08:00 3220812128-R1 RIP/7/DBG: 6: 12176: RIP 1: Sendi
ng v1 response on GigabitEthernet0/0/0 from 10.20.30.177 with 3 RTEs
May 15 2024 13:46:45.790.2-08:00 3220812128-R1 RIP/7/DBG: 6: 12227: RIP 1: Sendi
ng response on interface GigabitEthernet0/0/0 from 10.20.30.177 to 255.255.255.2
55
May 15 2024 13:46:45.790.3-08:00 3220812128-R1 RIP/7/DBG: 6: 12247: Packet: Vers
ion 1, Cmd response, Length 64
May 15 2024 13:46:45.790.4-08:00 3220812128-R1 RIP/7/DBG: 6: 12296: Dest 0.0.0.0
, Cost 1
May 15 2024 13:46:45.790.5-08:00 3220812128-R1 RIP/7/DBG: 6: 12296: Dest 10.20.3
0.176, Cost 1
May 15 2024 13:46:45.790.6-08:00 3220812128-R1 RIP/7/DBG: 6: 12296: Dest 10.0.0.
0, Cost 1
May 15 2024 13:46:40.770.1-08:00 3220812128-R1 RIP/7/DBG: 6: 12185: RIP 1: Recei
ving v2 response on GigabitEthernet0/0/0 from 10.20.30.178 with 2 RTEs
May 15 2024 13:46:40.770.2-08:00 3220812128-R1 RIP/7/DBG: 6: 12236: RIP 1: Recei
ve response from 10.20.30.178 on GigabitEthernet0/0/0
May 15 2024 13:46:40.770.3-08:00 3220812128-R1 RIP/7/DBG: 6: 12247: Packet: Vers
ion 2, Cmd response, Length 44
May 15 2024 13:46:40.770.4-08:00 3220812128-R1 RIP/7/DBG: 6: 12315: Dest 10.20.3
0.0/26, Nexthop 0.0.0.0, Cost 1, Tag 0
May 15 2024 13:46:40.770.5-08:00 3220812128-R1 RIP/7/DBG: 6: 12315: Dest 10.20.3
0.64/26, Nexthop 0.0.0.0, Cost 1, Tag 0
May 15 2024 13:46:40.770.6-08:00 3220812128-R1 RIP/7/DBG: 6: 2500: RIP 1: Ignori
ng packet. This version is not configured.
May 15 2024 13:46:44.590.1-08:00 3220812128-R1 RIP/7/DBG: 25: 3741: RIP 1: Perio
dic timer expired for interface GigabitEthernet0/0/0
```

而 RIPV2 是使用组播地址去发送

```
May 15 2024 13:40:30.980.1-08:00 3220812128-R1 RIP/7/DBG: 6: 12176: RIP 1: Sendi
ng v2 response on GigabitEthernet0/0/0 from 10.20.30.177 with 4 RTEs
May 15 2024 13:40:30.980.2-08:00 3220812128-R1 RIP/7/DBG: 6: 12227: RIP 1: Sendi
ng response on interface GigabitEthernet0/0/0 from 10.20.30.177 to 224.0.0.9
May 15 2024 13:40:30.980.3-08:00 3220812128-R1 RIP/7/DBG: 6: 12247: Packet: Vers
ion 2, Cmd response, Length 84
May 15 2024 13:40:30.980.4-08:00 3220812128-R1 RIP/7/DBG: 6: 12315: Dest 0.0.0.0
/0, Nexthop 0.0.0.0, Cost 1, Tag 0
May 15 2024 13:40:30.980.5-08:00 3220812128-R1 RIP/7/DBG: 6: 12315: Dest 10.20.3
0.128/27, Nexthop 0.0.0.0, Cost 1, Tag 0
May 15 2024 13:40:30.980.6-08:00 3220812128-R1 RIP/7/DBG: 6: 12315: Dest 10.20.3
0.160/28, Nexthop 0.0.0.0, Cost 1, Tag 0
May 15 2024 13:40:30.980.7-08:00 3220812128-R1 RIP/7/DBG: 6: 12315: Dest 10.20.3
0.176/30, Nexthop 0.0.0.0, Cost 16, Tag 0
May 15 2024 13:41:44.600.1-08:00 3220812128-R1 RIP/7/DBG: 6: 12185: RIP 1: Recei
ving v2 response on GigabitEthernet0/0/0 from 10.20.30.178 with 3 RTEs
May 15 2024 13:41:44.600.2-08:00 3220812128-R1 RIP/7/DBG: 6: 12236: RIP 1: Recei
ve response from 10.20.30.178 on GigabitEthernet0/0/0
May 15 2024 13:41:44.600.3-08:00 3220812128-R1 RIP/7/DBG: 6: 12247: Packet: Vers
ion 2, Cmd response, Length 64
May 15 2024 13:41:44.600.4-08:00 3220812128-R1 RIP/7/DBG: 6: 12315: Dest 10.20.3
0.0/26, Nexthop 0.0.0.0, Cost 1, Tag 0
May 15 2024 13:41:44.600.5-08:00 3220812128-R1 RIP/7/DBG: 6: 12315: Dest 10.20.3
0.64/26, Nexthop 0.0.0.0, Cost 1, Tag 0
May 15 2024 13:41:44.600.6-08:00 3220812128-R1 RIP/7/DBG: 6: 12315: Dest 10.20.3
0.176/30, Nexthop 0.0.0.0, Cost 16, Tag 0
May 15 2024 13:41:54.200.1-08:00 3220812128-R1 RIP/7/DBG: 25: 3741: RIP 1: Perio
dic timer expired for interface GigabitEthernet0/0/0
```

实验总结：

通过本次实验，我了解了新的一种动态路由协议 rip，通过配置实验，我知道了 ripv1 和 ripv2 的区别。同时也清楚了动态路由协议和静态路由协议的区别，以及了解了 ripv1 和 ripv2 是如何去添加路由条目的，还知道了如何用反向发布去把路由表发送给其他主机。

命令附录：

1） 启动RIP进程1

[Huawei]rip 1

2) 宣告参与更新接口所在主类网络号

[Huawei-rip-1]network 10.0.0.0

3）路由debug

< Huawei >debugging rip 1　开启RIP调试

< Huawei >dis debugging　　查看当前的调测信息

< Huawei >terminal debugging debug信息在终端屏幕显示

< Huawei >undo debugging rip 1　或者

< Huawei >undo debugging all　　关闭调试

4）反向发布默认路由

[Huawei -rip-1]default-route originate

jixr